

K1: Clear and Comprehensive Summary

Looking back on our group's engineering design process, we implemented a thorough process of defining the problem, hypothesizing ideas, selecting a design, building our prototype, testing and evaluating, and looking further for redesign.

We started off by identifying a problem in our community. We found that at our local elementary school, teachers have been having trouble teaching students with learning disabilities how to wash their hands. The teachers found that the process of taking a student to the nearest sink could be too distracting to the learning environment. The teachers requested if we could find a way we could bring this teaching environment to the students. Our group quickly came to the conclusion that we would need to design and construct a portable sink for this classroom. With this in mind, we researched popular producers of portable sinks, and analyzed their pros and cons. We found sinks with a variety of materials sizes, and flow methods. From this research we were able to gain some guidance for possible designs. After identifying the general goal, we looked towards design criteria and requirements. We set our main focus to constructing a realistic sink that could allow a teacher to accurately demonstrate how to wash hands and brush teeth. We also valued cleanliness, portability, and storage, among other metrics. Next, we set targets for our sink's abilities. We used our current knowledge of sinks, the criteria for our solution, and other information to determine our sink's goals. We set goals for things such as weight, flow rate, and size.

Now that we have set our targets, our group was ready to begin designing our sink. For element D, our group split up to encourage as much creativity as possible. Each group member was assigned to make one design based off of each of our ideas. Each member kept in mind our sink criteria and limitations while designing. The biggest challenge may have been keeping our limited building abilities in mind. Obviously, we couldn't simply spawn any sink we could design, so we had to be realistic with ourselves. After making our designs, we consulted our decision matrix. We made this decision matrix together, based on our original sink criteria. Next, we graded each design. The simplicity of our first design was probably the biggest factor determining our choice. After this, we started our preparation for constructing our sink. First, we made a list of the supplies we would need to purchase. We then highlighted STEM aspects we would utilize in our design. We also identified experts we were in contact with whom could advise us in our construction process. Next, it was extremely important that we identified possible hazards we may run into throughout our construction process. The use of water and our cooperation with young students added an extra layer of caution to our process. We looked into spillage, cleanliness, and stability, among other key aspects.

After these steps were completed, our group was ready to start construction. We followed the design as closely as possible, but ran into some minor obstacles. For example, we did not plan out how we would cut out our plastic box to make room for our sink. However, we were able to overcome these obstacles. One change we made was leaving the back of the sink cabinet open. This reduced the weight and allowed for easier use by teachers. This would not affect the

student's view. We also decided to stain the wood of our cabinet. In addition, we made some changes to the sink spigot function from the original paper design. Throughout this process, we highlighted current struggles and concerns to help with the clarity of our ideas. We also decided to make use of the open space with a mirror.

Now that our sink was built, we looked towards testing the sink. We started by planning out how we would measure each metric. Metrics like flow rate and flow length were pretty simple to test. Some things like visual appearance needed consulting from an expert. At first, we did not fully think out how we would test our goals. For example, we had to go to the nurse's office to weigh our sink. However, we were able to grade all of our metrics in the end. We also prioritized these gradings, which could help if we decide to make another improved version of our sink.

K2: Value Judgment

Overall, the sink design had many positive and many negative ideas, prototypes, build techniques, conversations, and results. For the group, we had mixes of expectations as to the look, design, quality, and function of the sink itself. During the designing phase, we spent long periods of time working through CAD via OnShape. While we were able to create an accurate idea of our prototype, the prototype took a long time to develop as only one of our members was familiar with OnShape designing. Further, another issue that cost a major loss in time, and a longer build-time was the disagreements and confusion of how the group wanted to go about manufacturing each piece of the project. During the build phase, we ran into issues, which became stretched out, as we found ourselves going in circles when discussing solutions. Another issue we had related to the group as a whole was our inability to focus on the task set ahead of us to do each class. Due to the complexity of the build and our lack of ability to make an adequate plan of action, there was not a healthy pace that we kept in the building process. This also caused many delays and extended the process of the project. Also, another issue was the inconsistency of professional finishes on pieces. While this issue may be subjective to each teammate, there were pieces that were not built to the standards we had hoped for. For example, one piece that was not finished perfectly was the front wood panel. This piece at certain viewpoints reveals that the front board was slightly bigger than the face it needed to cover, revealing slight overhang. While this has no real effect for the student or the teacher, it may cause some visual displeasure if one is bothered by small details as such.

Despite the many flaws and mishaps over the process of designing and building, there were many positives that occurred, that set us up to finish the project satisfied with the results. One positive of the project was the amount of materials that were able to be used from our in-school inventory. This mitigated the long stretches of waiting for materials to come in to continue building. Further, our access to Amazon Prime allowed us to get all our out of school materials in less than a 3-day span. Another positive of the process of completing the project was our ability to assess past attempts of the intended product. Our school had multiple groups in the past attempt to solve this problem, however due to many factors, they were unable to complete

the project successfully. On top of that, while this may be assumed, another positive was that we did complete the project unlike past groups.

K3: Inclusion of Lessons Learned

Throughout our project, we experienced many setbacks. The number one thing we struggled with was staying on task and figuring out what to do next. Throughout the engineering design process, we struggled most with the planning portion. Our research and ideas went very smoothly and things started to slip up around the planning portion. We also did not get our supplies for an extended period of time and were not placing orders for parts soon enough. We did not think ahead to what could happen if something were to break or fall off, causing more issues when something like that would happen. Based on our experience in this project, we feel as though other designers should prioritize the planning phase and stick to it. Other designers should also be aware of focusing too much time on one small thing rather than focusing on the big picture. A lot of time was spent debating the faucet and handles when that time could have been used to start the physical building of our project, buying more materials, and working more on the paperwork for the project. Across the whole engineering design process, we learned that the questions we ask and the things we need to figure out need to be specific but not too specific; basically enough to base a project on. Research is integral to the entire project and without good research, the project will suffer. There were many times where we needed to figure things out that we had never accounted for in the beginning, like if filament in a 3d printer is FDA approved. We seemed to be a little close minded when it came to our sink designs, all of them really looked the same and if there was more variety, we may have had more of a different outlook on the project. As previously mentioned, the planning stage was where things started to fall apart for the group. Our plans were not clear enough and did not include several key details, so we needed to reevaluate, pushing the project back. Creating a prototype was challenging as well as there were some innergroup dynamic struggles and there would tend to be only one person working on the physical making of the project for a majority of the time. The testing portion was not bad, as we did have enough information on how and what to measure and test. In terms of improving the project, where the group is at now it does not seem that we will get to that part. Our first prototype seems to be our final prototype for the stakeholders.